

Seaside **SOLDIERS** *Report*



ENGR 7A: Introduction to Engineering

University of California, Irvine
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Executive Summary

From the beginning of the project, the first task we set out to do was to establish roles and group meetings. During our first meeting, we identified the skills of each team member, as roles were assigned each team member would contribute and help one another on completing tasks. During the first meeting, we also sketched out the position of hardware and the general structure of the rover. Alongside sketching out the structure rover, we identified the constraints of the role such as the size, budget, and safety. After the first group meeting, the first iteration of steering and rover structure was completed. This first iteration was used to get an understanding of the type of steering we wanted to design and use.

On our second meeting, we continued to sketch out the design of the rover, sourced the rover parts and created a Gantt Chart. During our meeting we also discussed the structure of the rover should be Italian poplar wood as it was lightweight but durable. Using the Gantt chart gave us a timeline and a schedule on what we should complete on that specific day. The Gantt chart helped us stay organized and prioritize the tasks that need to be completed.

After our third meeting, the steering “prototype” in Solidworks was completed. We decided to use Ackerman steering for our rover as it is more efficient at controlling the vehicle's turning, especially at low speeds. Having completed the prototype, Solidworks gave us a visualization on the rover dimensions and the chassis shape. We were also able to get a better understanding of the hardware positions on the chassis. By our third meeting, we were able to complete the purchase order form to get started on fabrication, from the purchase order form we were able to get a better understanding of how our budget will be spent.

After receiving our materials, fabrication could be started. During the lab, we were able to complete the rover chassis and complete the axle housing. We were also able to complete the electronics and pair the receiver and transmitter. Our 3D printed steering was also completed, though problems with the servo arm link occurred. We determined that the servo arm was loose and unable to turn the steering. Thus, we created an improvised version of the servo arm link. We created a notch that would surround the steering link, making the steering functional. After the full integration of the electronics to the rover, we were able to start testing on the rover.

During our first run through the course, we identified two different problems. The first problem was the rover was too low to the ground, making it difficult to go over the ramp and cones. The solution to this problem was approaching the ramp with an angle at the peak, then accelerating at the peak due to the low body structure. The second problem was the servo arm link broke off. Thus, leading back to redesigning and enforcing the structure of the steering system and servo arm. After many trials and tribulations and different iterations, we were able to do a speed test and go through the course easily.

Introduction and Problem Definition

Our objective was to build a working rover over the course of 10 weeks ensuring that it could successfully complete an obstacle course consisting of several cones and a ramp. Our team wanted to ensure the rover met all the safety requirements, had a sleek design, had a functional steering mechanism, and had enough clearance to complete the obstacle course without getting stuck.

Technical Review / Background

Rovers were first invented as a method of exploring spaces where humans weren't able to reach. One of the earliest iterations of the rover was the Sojourner, which was a part of the Pathfinder mission. It was pivotal in the very first exploration of Mars, where its cameras first captured images of pebbles and cobbles which led to the discovery of water on Mars which created the possibility of life on other planets in our very own solar system (Raith M., 2024).

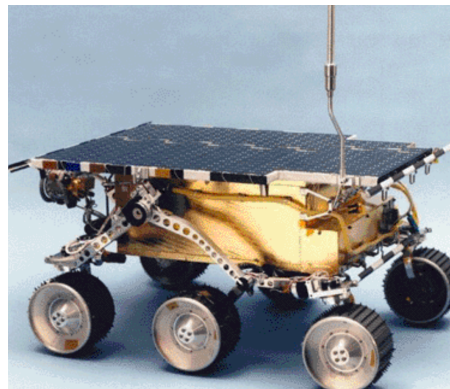


Figure 1: Sojourner rover

Rovers were also used to explore heavy and hard to reach areas as they were built to be more durable than regular vehicles. For example the U.S. military's Special OPs division uses the Ranger Special Operations vehicle, the SOV, as a lighter and more convenient way to transport troops through rough terrain and is also used for light attacks and reconnaissance. It is so light it can be transported using a military helicopter (*Special Operations Vehicles, 2024*). In more recent years, technological advancements have made rovers more effective in completing certain tasks, an example of this is NASA's Perseverance which was designed to collect samples of Mars's surface and store them in tubes. The rover is equipped with advanced technology like Raman Spectrometers that shine UV light onto rocks to determine their properties (*The Planetary Society 2021*).

While our rover wasn't designed to accomplish life changing discoveries, we were able to design a rover that was able to run a track and steer on its own with the help of a wireless controller. We decided to design our rover's chassis and overall build after F1 race cars, they prioritized a sleek and slim design and were lower to the ground in order to take advantage of aerodynamics and acceleration forces (*Red Bull Racing, 2022*). We decided on a tapered silhouette, where the front was smaller and less wider than the body to allow for aerodynamics and to have an advantage when turning as slimmer body's can typically make sharper turns. Prioritizing aerodynamics where we

could was also a priority as lighter rovers could accelerate faster, which is why we used 6mm Italian Poplar wood so our rover would be sturdy enough to withstand impact and hold hardware but light enough to have a fast acceleration. We also placed the electronics in such a way that our rover was weighed down evenly so our rover wouldn't tip over. We also decided to smoothen the edges of our rover to allow for better airflow around it. We additionally decided to use Ackerman steering for our rover because it enabled us to design a more simpler steering system that was easy to 3D print, which allowed our rover to have a smaller build as well as allow us to make sharper turns.

Design Requirements

Size requirements

- (a) Overall width must be smaller than 10 inches.
- (b) Overall length must be smaller than 16 inches.

Structure requirements

- (a) Your team will design and fabricate the rover structure.
- (b) Using an off the shelf chassis is not allowed.
- (c) Off the shelf steering mechanisms or individual linkage components are not allowed.
- (d) You are allowed to use raw materials purchased externally with the permission of your TA or technical staff.
- (e) 3D printing of the steering mechanism and any associated attachments is allowed, but must be counted toward your budget.

Power and propulsion requirements

- (a) The use of internal combustion engines (gas and glow fuel engines) is prohibited.
- (b) Rover must be able to operate for at least 10 minutes without replenishing/recharging its energy source.
- (c) On/off switch is required and must be easily accessible.
- (d) Battery must be easily removable.

Safety requirements

- (a) Must include a gear guard to protect against any contact with the gears when in motion.
- (b) No protruding sharp objects. All sharp corners must be filed or sanded down.
- (c) All wires must be neat and away from any moving parts. This includes, rotating axles, motor shafts, servos, linkages, and gears.
- (d) All wires and connectors are to be completely covered and insulated.

Cost requirements

- (a) Total as-built replacement cost of the rover must be less than \$300 including the costs

associated with 3D printing and any external parts you may choose to purchase.

(b) The cost must be broken down into a Bill of Materials (BOM)/Parts List, in which the Fair Market Value (FMV) of each component must be listed.

Design Description

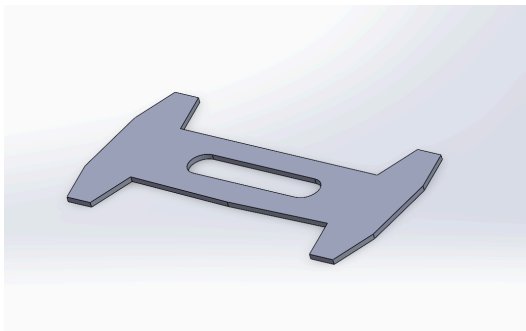
Summary of The Design

For our rover, we wanted to make it as durable as possible, while still having good acceleration. Because of this, we chose to make our chassis with Italian Poplar Plywood. A durable plywood which is easy to work with, and light in weight allowing for good acceleration.

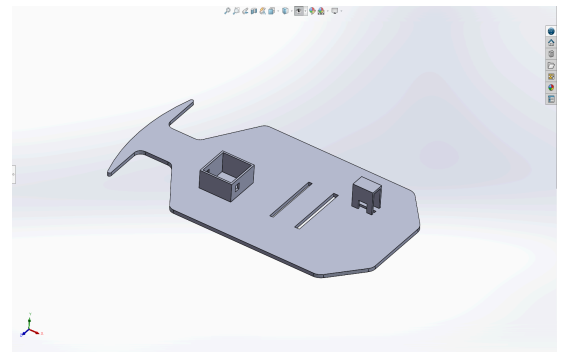
We chose the 85mm wheels because they require less force to rotate than the larger ones, resulting in faster acceleration.

Design Details

First Iteration: Chassis



Second Iteration: Chassis

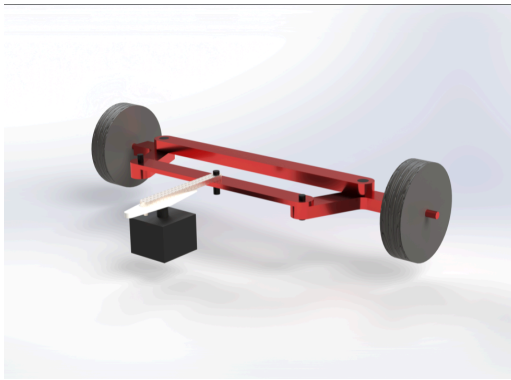


Third Iteration: Chassis

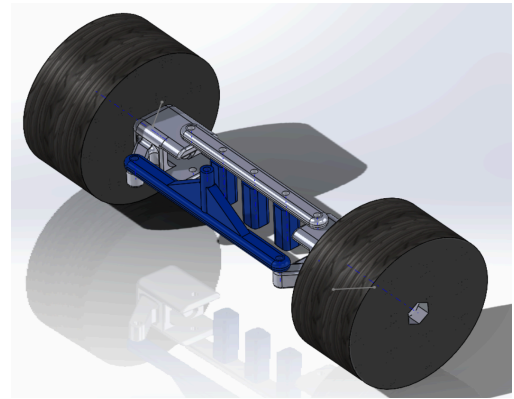
We first wanted to go with a sleek design as shown by our first iteration on the left, so we decided with wider ends and a thinner middle to account for the battery and parts. However, issues came up with durability and the fact that there may not have been enough clearance for the parts so we decided to redesign the chassis and keep one end of the rover thinner so it could hold the steering and have a wider body in order to hold the hardware and electronics. This is shown in the second iteration on the right. In the third iteration, we decided to round the edges so our rover would look

smoother, and to account for safety issues with sharp corners. We also decided to add a hole in the back of the chassis so the gear could be placed without any issues, and two slits in the middle of the chassis so the battery would have velcro straps wrapped around it. This would solve our balance issues as the battery, being the heaviest part of our rover would go in the middle and would ensure our rover would not tip over while driving.

First Iteration: Steering System

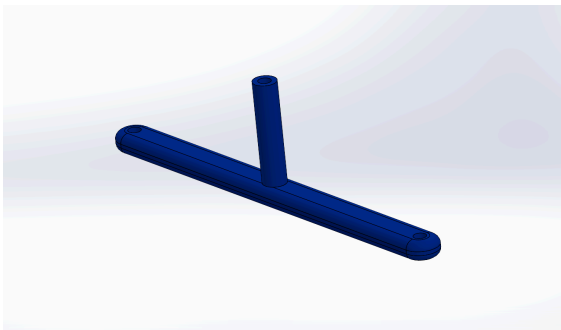


Second Iteration: Steering System

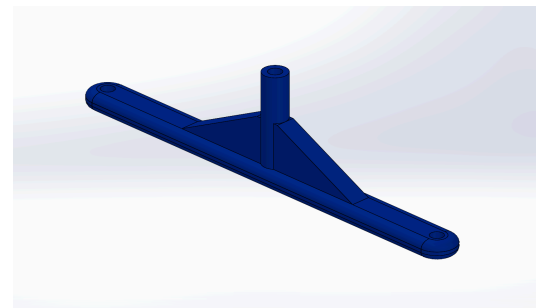


In regards to our steering mechanism, we also decided to use Ackerman steering in order to make sharper turns. Our first iteration shown on the left, was an earlier design where the servo arm was directly attached to a bar in the middle and which would rotate with the servo. However, because of clearance issues and the fact that the servo was taller than the steering, we had to redesign the system. In our newer design, we added 3 blue pillars so that we could screw in the steering system to our chassis. They would offer a stable form of attachment instead of gluing them on. We also changed where the servo arm would be. It would go on top of the actual steering bar and connect to the servo when it's upright. We also refined the design and added axle mounts to support the wheels.

First Iteration: Link Arm

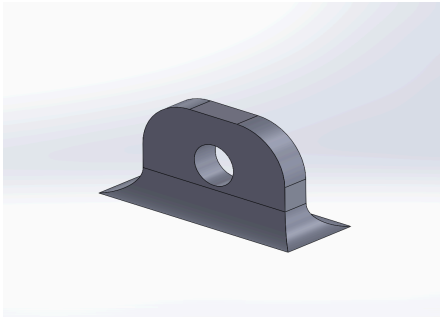


Second Iteration: Link Arm

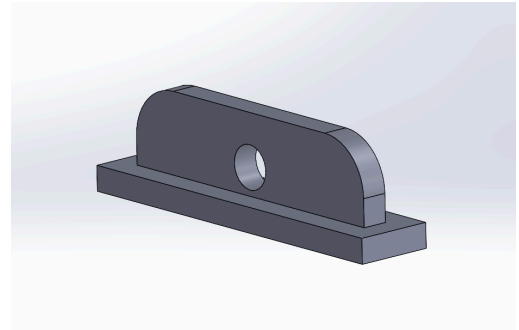


The first iteration snapped while test-driving the rover so we added supports to the side in the second iteration which prevented it from snapping in future tests.

First Iteration: Axle Housing



Second Iteration: Axle Housing



The first iteration was too thin and did not have enough surface area on the bottom. In the second iteration, we made the overall part wider to increase surface area and the bottom part thicker to increase strength.

Below are three different views of our rover's final design along with the center of mass.

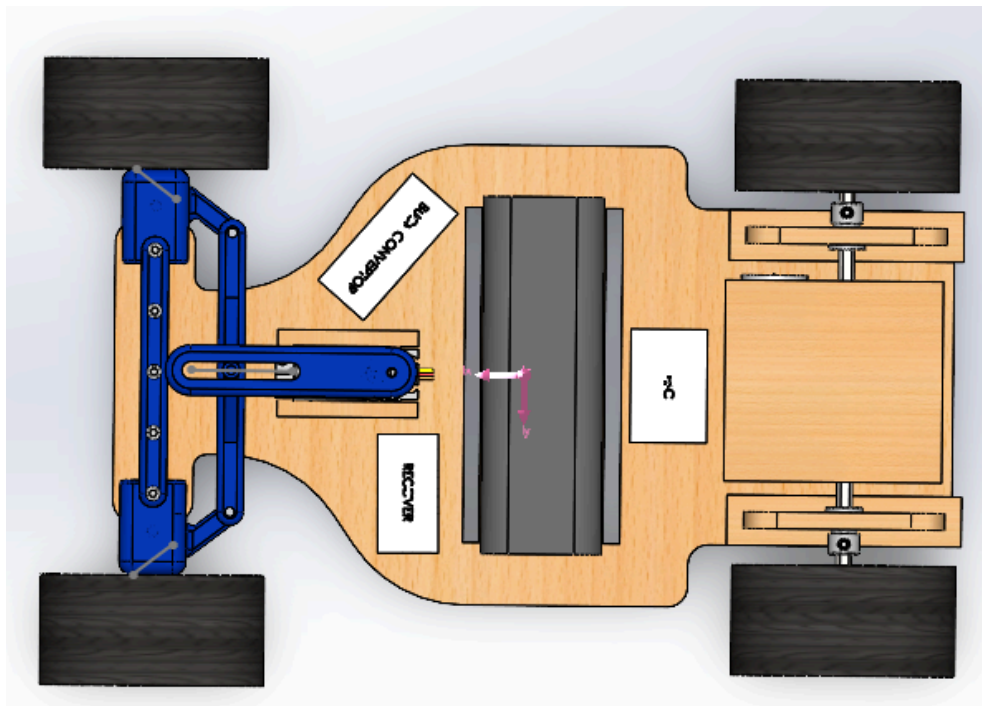
Center of mass: (inches)

$$X = -4.58$$

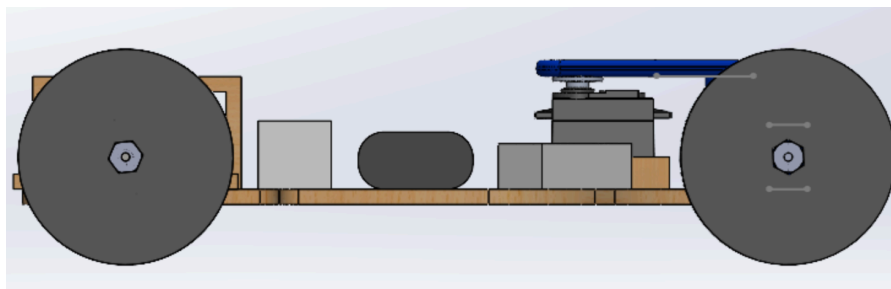
$$Y = 3.03$$

$$Z = 7.24$$

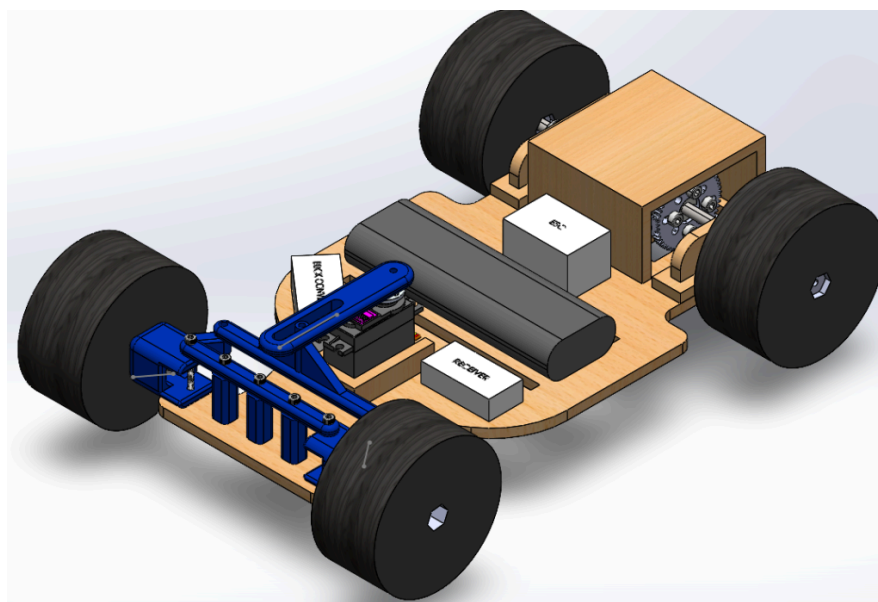
Top View of Design



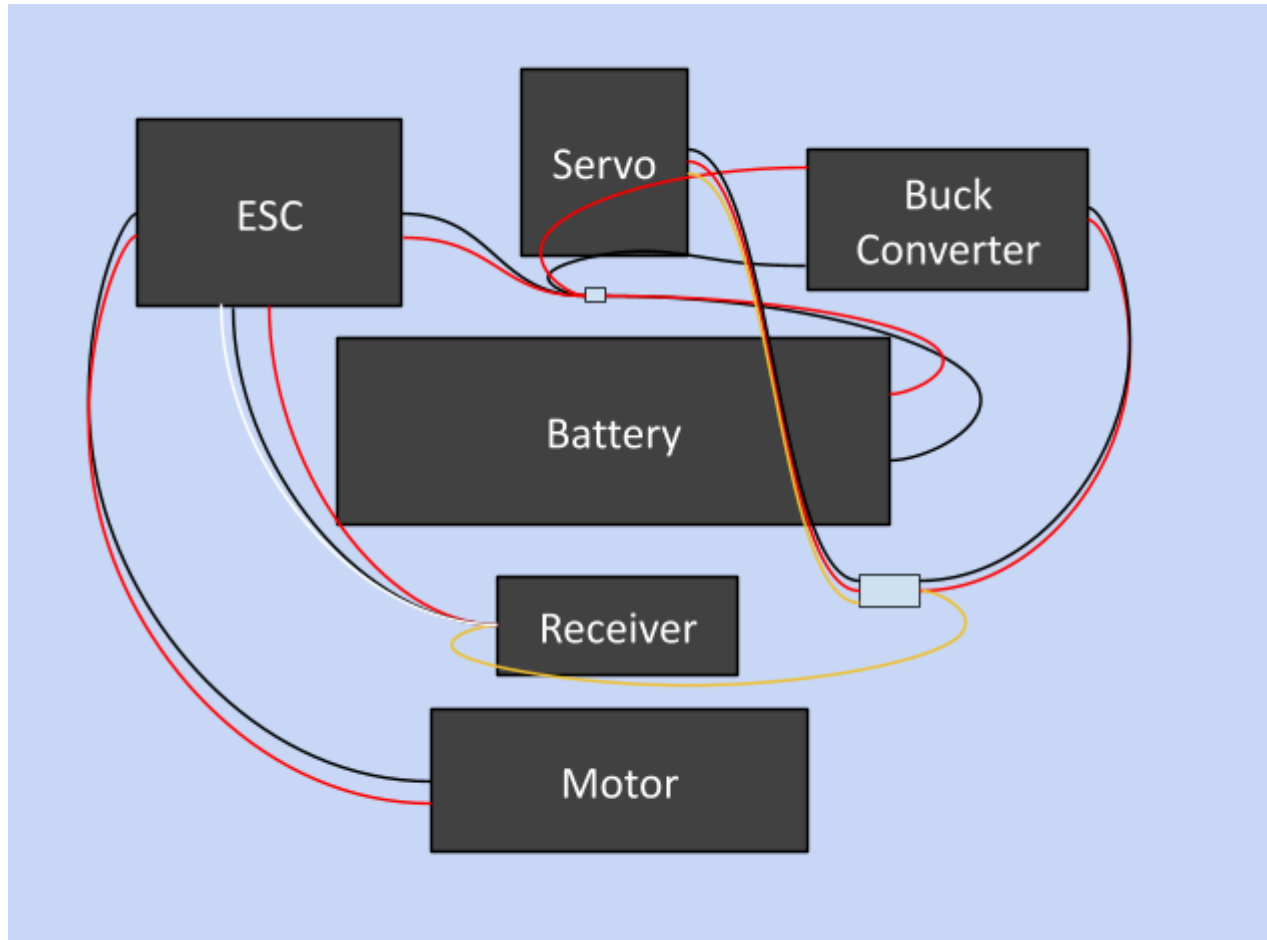
Side view of final design:



Isometric View of Final Design:



Wiring Diagram



Displayed here is the diagram of our electronics as well as how the wires were organized in the final design of our rover. We decided to switch the places of the ESC and buck converter since the wires kept getting tangled and our team had trouble taping the wires down, so they wouldn't stick out too much. Even though there is a wire going over the battery, our rover's chassis size and wire length did not allow us to move it to the side of the battery.

Action Item Report

Task Assignment

Team Formation

Within the first week of meeting with our team and getting to know each other, our first task assignment was to come up with a team name, as well as divide the project responsibilities among our members according to each of our strengths and interests. Many of our team members were interested in multiple different aspects of the project, so we ultimately decided that we could all participate in each subteam in order for us to gain experience individually.

Rover Design

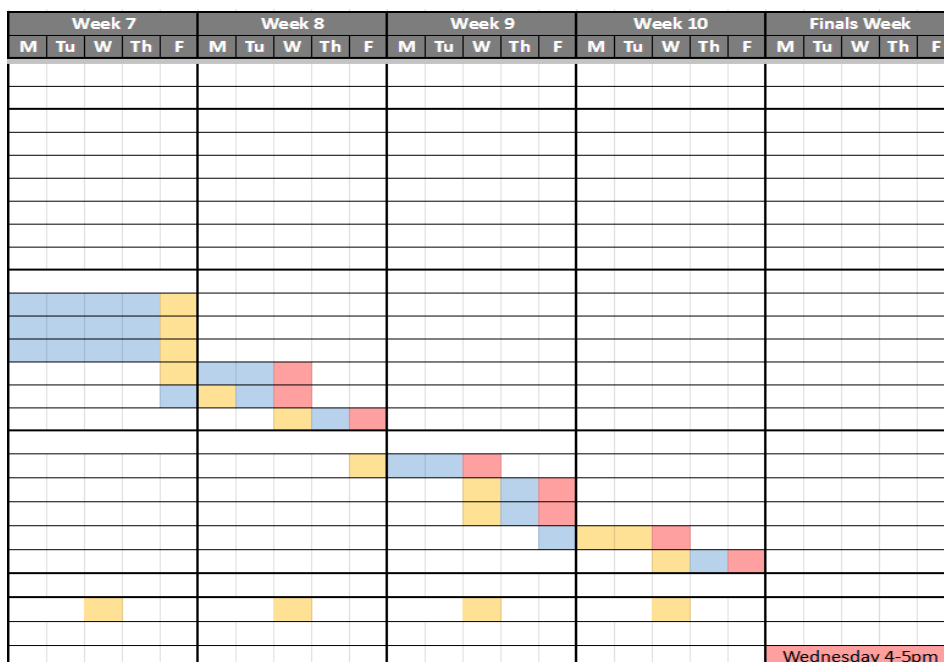
From week 3 to week 6 our primary task was to complete the rover design on SolidWorks, as well as finalize the specifications of our rover. During the earlier weeks, we met as a team to brainstorm our overall rover layout which consisted of online research regarding chassis design, gear ratio, parallel and Ackermann steering, and wheel sizing. After deciding which parts and materials were best suited for our particular ideal rover design, we then finalized our Purchase Order Form, ensuring that we were working within our budget. In the latter weeks, our team focused heavily on our SolidWorks design. Our Ackermann steering design took up most of our time, as the rest of our chassis design was dependent on the dimensions of our steering mechanism. We also completed SolidWorks drawings in order to better prepare for our structural fabrication.

Structure Fabrication

After completing our CAD on SolidWorks, we used our blueprint drawings to start our fabrication process. We began with our chassis design, which we made out of Italian poplar plywood. Using the dimensions that we finalized on SolidWorks, we sketched out our chassis onto the wood board that we ordered. From there, using the machines and tools that were provided in the lab, we were able to cut out our chassis with accurate dimensions that were within the requirements. Once we had our chassis, the remaining fabrication consisted of attaching the components using fasteners as well as attaching our 3D-printed steering mechanism. One important factor that we had to keep in mind while attaching our components was being able to easily and safely wire the electronics across our rover.

Electrical System

In the final weeks, our primary task was wiring the components together and making sure each component functioned properly. We initially had some challenges with organizing the wires within our design, ensuring that they would be safe during the driving process. We eventually were able to tape the wires onto our chassis neatly and performed test drives to make sure they were secure. Another obstacle that we encountered was the fact that our gears weren't fitting well together due to the slightly inaccurate fastening of the motor bracket. We had to remeasure where to drill holes and were eventually able to fasten the motor in a place where the pinion gear fit perfectly with the driven gear which was attached to the rear axel. By the end of week 8, our rover was fully complete and ready for testing



Evaluation

Calculations

Weight of the Vehicle: 1156.8 grams

Turning radius: 18.339 inches

Center of Mass: (5.045 in, 3.405 in)

Average Velocity: 3.91 ft/s

Acceleration: 0.465 ft/s^2

Predicted Drive Time: 0.7 hrs

The mass of our rover is 1,156.8 grams, with the center of mass of the rover being 5.045 inches to the right of and 3.405 inches above the bottom left wheel. The turning radius, calculated using the provided calculator, came out to be 18.339 inches with the wheelbase being 0.25 meters and the turning angle being 32.46 degrees. The average velocity of the rover is 3.91 feet per second and the acceleration is 0.465 feet per second per second. The predicted drive time of our rover is 0.7 hours or 42.5 minutes; however, our rover was able to run for longer than the predicted time.

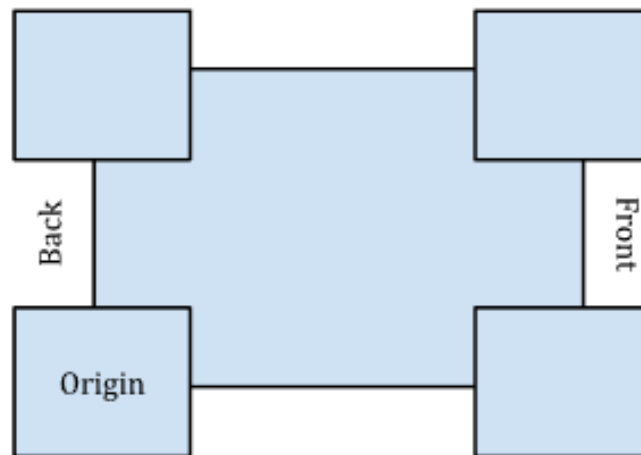
Center of Mass

$$m = 356.4\text{g} = 0.3564 \text{ kg}$$

$$W = 0.3564(9.8) = 3.493 \text{ N}$$

$$m = 207.8\text{g} = 0.2078 \text{ kg}$$

$$W = 0.2078(9.8) = 2.036 \text{ N}$$



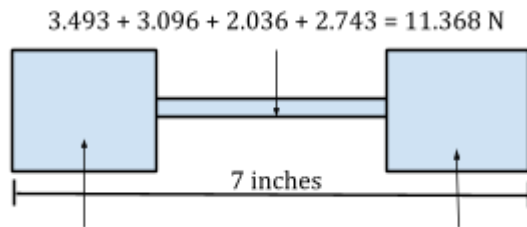
$$m = 315.9\text{g} = 0.3159 \text{ kg}$$

$$W = 0.3159(9.8) = 3.096\text{N}$$

$$m = 279.9\text{g} = 0.2799 \text{ kg}$$

$$W = 0.2799(9.8) = 2.743\text{N}$$

Front View

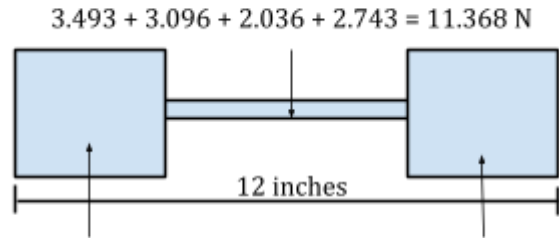


$$W = 2.743 + 3.096 = 5.839 \text{ N}$$

$$W = 2.036 + 3.493 = 5.529 \text{ N}$$

$$\begin{aligned}\Sigma M_{y, \text{origin}} &= (5.839)(0) + (5.529)(7) - (11.368)(y) = 0 \\ 38.703 &= 11.368y \\ y &= 3.405 \text{ inches}\end{aligned}$$

Side View



$$W = 3.493 + 3.096 = 6.589 \text{ N}$$

$$W = 2.036 + 2.743 = 4.779 \text{ N}$$

$$\begin{aligned}\Sigma M_{y, \text{origin}} &= (6.589)(0) + (4.779)(12) - (11.368)(x) = 0 \\ 57.348 &= 11.368x \\ x &= 5.045 \text{ inches}\end{aligned}$$

Center of mass of the rover from the origin:
(5.045 in, 3.405 in)

We calculated the rover by first measuring the masses of each side of the rover using the scales in the lab. After getting each mass, we were able to calculate the weight on each of the wheels. To measure the center of mass in the x-direction, we added the weights that were aligned in the y-direction. Then, we multiplied each side by its distance from the bottom left wheel, which we took as the origin. We then divided it by the total weight of the rover to get the center of mass from the origin in the x-direction. To get the center of mass in the y-direction, we repeated the same process, but added the weights that were aligned in the x-direction.

Rover Average Velocity

Distance Travelled: 10 feet

Travel Time: 2.56 seconds

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{10 \text{ ft}}{2.56 \text{ s}} = 3.91 \text{ ft/s}$$

Rover Acceleration

$$\text{Initial Velocity} = \frac{1 \text{ ft}}{0.82 \text{ s}} = 1.19 \text{ ft/s}$$

$$\text{Final Velocity} = \frac{1 \text{ ft}}{0.42 \text{ s}} = 2.38 \text{ ft/s}$$

$$\text{Acceleration} = \frac{\text{Change in Velocity}}{\text{Time}} = \frac{2.38 \text{ ft/s} - 1.19 \text{ ft/s}}{2.56 \text{ s}} = 0.465 \text{ ft/s}^2$$

We calculated the rover's average velocity using the speed test track given. The length of the track is 10 feet with each foot marked by a line. The total time the rover took to complete the speed track was 2.56 seconds. The formula for velocity is displacement divided by time, so 10 ft divided by 2.56 seconds equals 3.91 feet per second.

To calculate the acceleration of the rover, we needed to find the initial and final velocity of the rover as the formula for acceleration is the change in velocity over time. We did this by calculating the velocity of the rover after completing the first foot of the track and the velocity of the rover after it completes the tenth foot of the track. The initial velocity of the rover came out to be 1.19 feet per second and the final velocity 2.38 feet per second. We then subtracted the initial velocity from the final velocity and divided it by the total time it took to complete the track. The acceleration came out to be 0.465 feet per second per second.

Predicted Drive Time

Capacity of battery is 3600 mAh $\rightarrow$ 3.6 Ah

Servo current (from Amazon): 5V $\rightarrow$ 3A

Motor current: (Initially) 12 voltage $\rightarrow$ 5A

But runs at 5 V $\rightarrow$ 2.08A

$$\text{Predicted Drive Time} = \frac{\text{Battery Capacity (Ah)}}{\text{Total Amperes}}$$

$$\text{Predicted Drive Time} = \frac{3.6 \text{ Ah}}{3 \text{ A} + 2.08 \text{ A}}$$

$$\text{Predicted Drive Time} = 0.7 \text{ hrs or } 42.5 \text{ mins}$$

To calculate the predicted drive time, we divided the battery capacity by the total amperes used by the motor and servo. The motor runs on 2.08 amperes and the servo uses 3 amperes. We got these values by finding the voltage to ampere conversion on the product description and using proportions to find the amperes for the voltage we set the motor and servo at. Our predicted drive time came out to be 0.7 hours or 42.5 minutes; however, our rover has proven to be able to drive for longer.

Mechanical Advantage, Stall Weight, Stall Torque

Mechanical Advantage

Pinion Gear (N_{input}) = 16 Teeth
 Driven Gear (N_{output}) = 52 Teeth

$$MA = \frac{N_{\text{output}}}{N_{\text{input}}} = \frac{52T}{16T} = 3.25$$

Stall Weight

Stall Mass = 1156.8 grams = 1.1568 kg
 Acceleration due to gravity (g) = 9.8 m/s^2

$$\text{Stall weight} = \text{stall mass} \times g = 1.1568 \text{ kg} \times 9.8 \text{ m/s}^2 = 11.33664 \text{ N}$$

Stall Torque

Stall Weight = 11.33664 N
 Pulley Radius = 0.0159 m

$$\tau_{\text{input}} = \text{stall weight} \times \text{pulley radius} = 11.33664 \text{ N} \times 0.0159 \text{ m} = 0.18025 \text{ Nm}$$

$$\tau_{\text{output}} = \tau_{\text{input}} \times MA = 0.18025 \text{ Nm} \times 3.25 = 0.58582 \text{ Nm}$$

Our group chose to use the 52 tooth gear as it was balanced in both angular velocity and torque. We calculated the mechanical advantage by dividing the number of teeth on the input gear, in this case the pinion gear, and the output gear. The pinion gear has 16 teeth and our output gear has 52 teeth, so 52 divided by 16 equals 3.25. We solved for the stall weight by multiplying the stall mass, 1.1568 kg, by the acceleration due to gravity, 9.8 m/s^2 ; the stall weight came out to be 11.337 N.

There were two stall torques to solve for: the input stall torque and the output stall torque. To solve for the input stall torque, we multiplied the stall weight, 11.337 N, by the pulley radius, 0.0159 m. For the output stall torque, we multiplied the input stall torque by the mechanical advantage. The input stall torque came out to be 0.18025 Nm, and the output stall torque came out to be 0.58582 Nm.

Test Plan

The main goal of testing our rover was to ensure its stability and efficacy when running. We first ran into problems in our initial design of the 3D printed steering in Solid works. Our axle mount would not be able to 3D print properly as the base of it had too much overhang and would snap if we used it. So, to solve this problem, we decided to make the axle mount out of wood instead. After 3D printing the parts, tested the steering by attaching it to the chassis, and then testing the steering range of motion as well as looking to see if the parts fit together.

After, we screwed on the motor and attached the electronics such as the ESC, receiver, and buck converted with double sided tape. Then, we connected the wiring to each of the parts, then tested to see if the motor was running, and if the electronics such as the ESC the servo responded to the electrical signals. Since all the electronics were in working order, we decided to finally attach the steering to our servo and test it by running our rover by driving it around to see our rover's steering range of motion with the servo and if the steering was working properly.

However, we shortly ran into problems with our steering mechanism. It ended up being extremely fragile and did not survive our first trial of the obstacle course. Additionally, the area where our servo was connected to the steering mechanism, the servo arm, was also very fragile as well. We decided to add extra supports to the main arm of our steering mechanism to ensure it's strength and endurance while running the obstacle track and switch out the servo arm from a screw, to a pin and slot arm made of wood. To test the efficacy of our new mechanism, we ran the rover over the obstacle course and found that our steering was working better. However, with that came another problem, our rover's height did not allow us to clear the ramp in the first few trials as its height was too low. Because of time constraints, we could not redo our rover's general structure to give it more clearance but a solution was found to this problem by driving the rover over the ramp at a certain angle, which allowed it to clear the ramp without any issues. So with this in mind, the rover was able to complete the course without any issues.

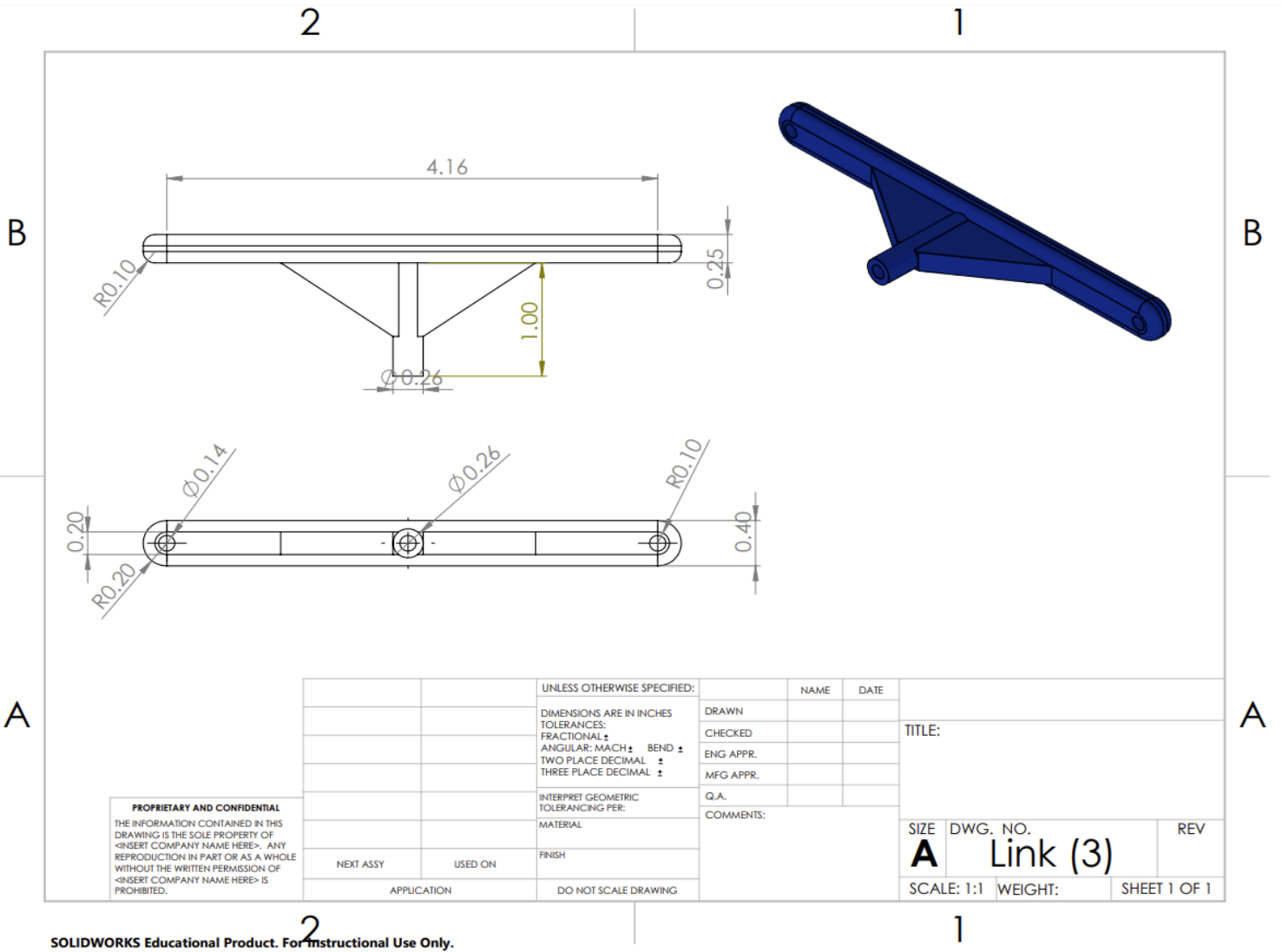
Results & Discussion

The most important criteria that we focused on in our rover was meeting the fabrication requirements, stability, agility and durability. To meet these criteria we placed our heavier parts such as the battery and the ESC towards the center of our rover so it would be more stable and not flip while driving.

We decided to add rounded edges to the rover in order to make it agile to account for aerodynamics and used Ackermann steering in order to make tighter turns. The steering was simple in order for it to be easily repaired in case it was damaged during testing. We also added extra supports to our 3D printed parts and .50 millimeters of clearance to that our parts fit together.

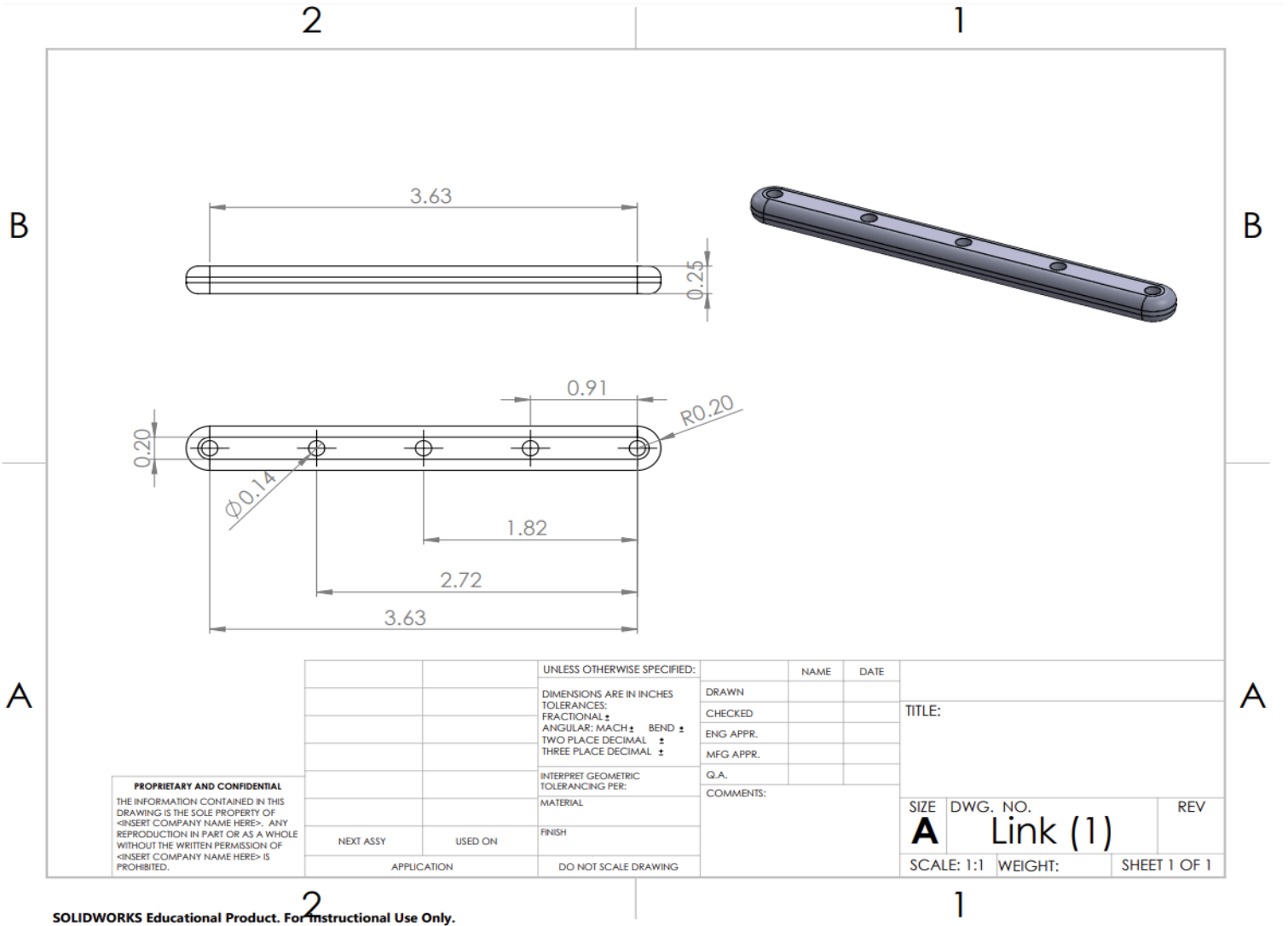
We had several problems while fabricating and finalizing our rover's design. First was the lack of durability of our 3D printed parts, and the measurements being off. And second, the rover's height did not allow it to clear the ramp easily. Next quarter, given the chance to build another rover, we would add extra supports to the 3D printed parts to prevent them from breaking, ensure that the measurements of the design are accurate and design the rover to have a higher clearance on its bottom side so it would be able to go over obstacles like ramps without much difficulty.

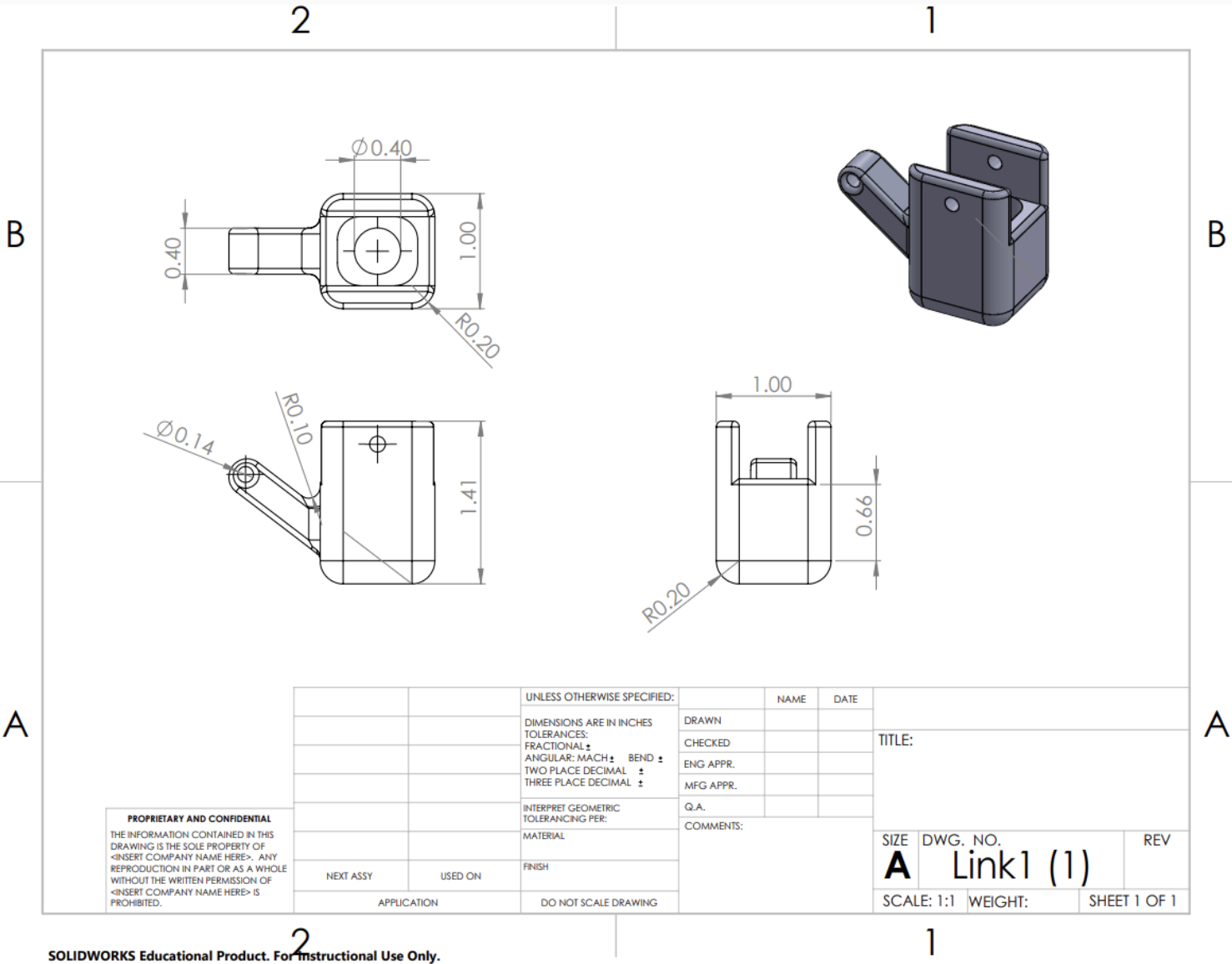
Appendix A: SOLIDWORKS Drawings

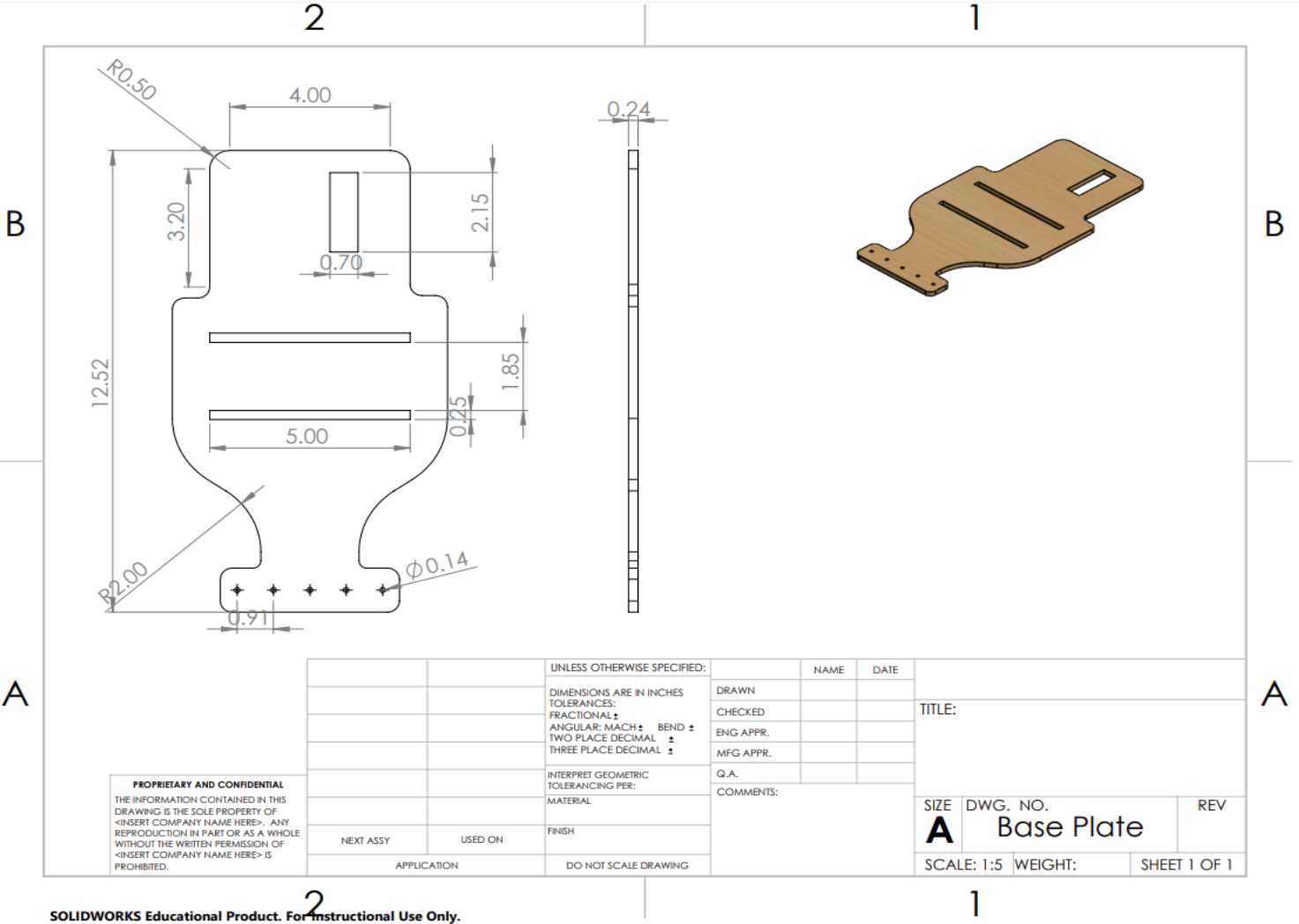


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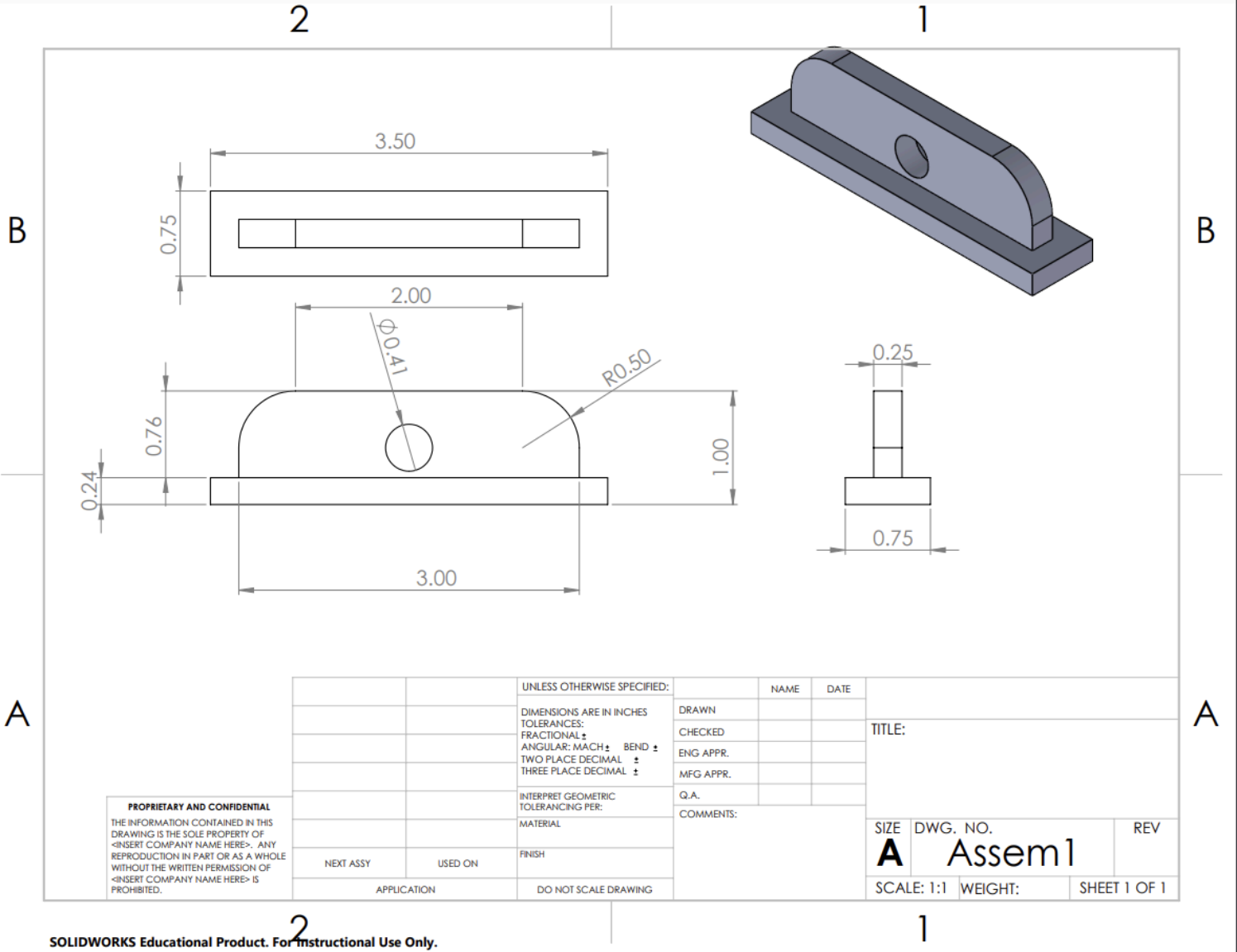
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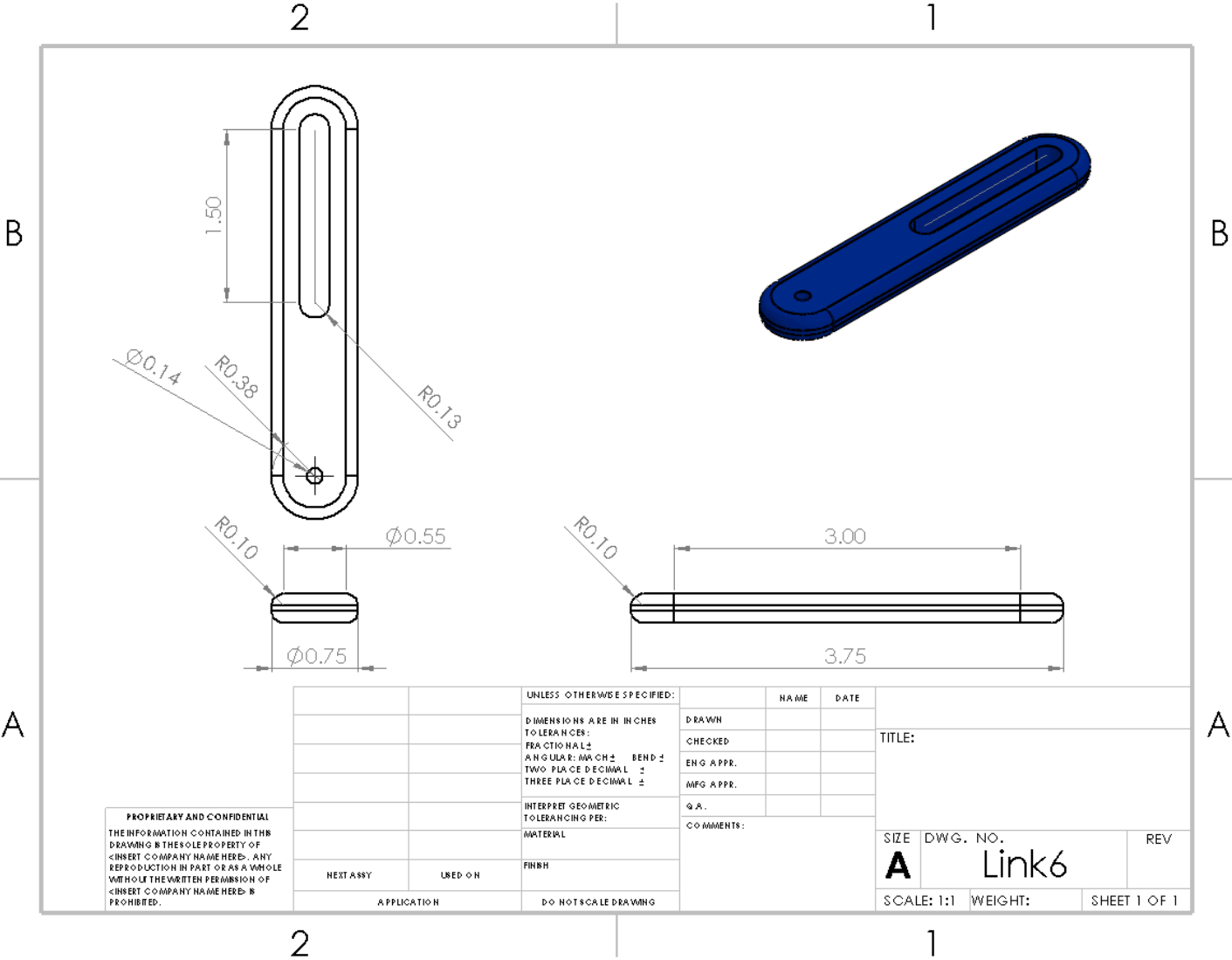
1

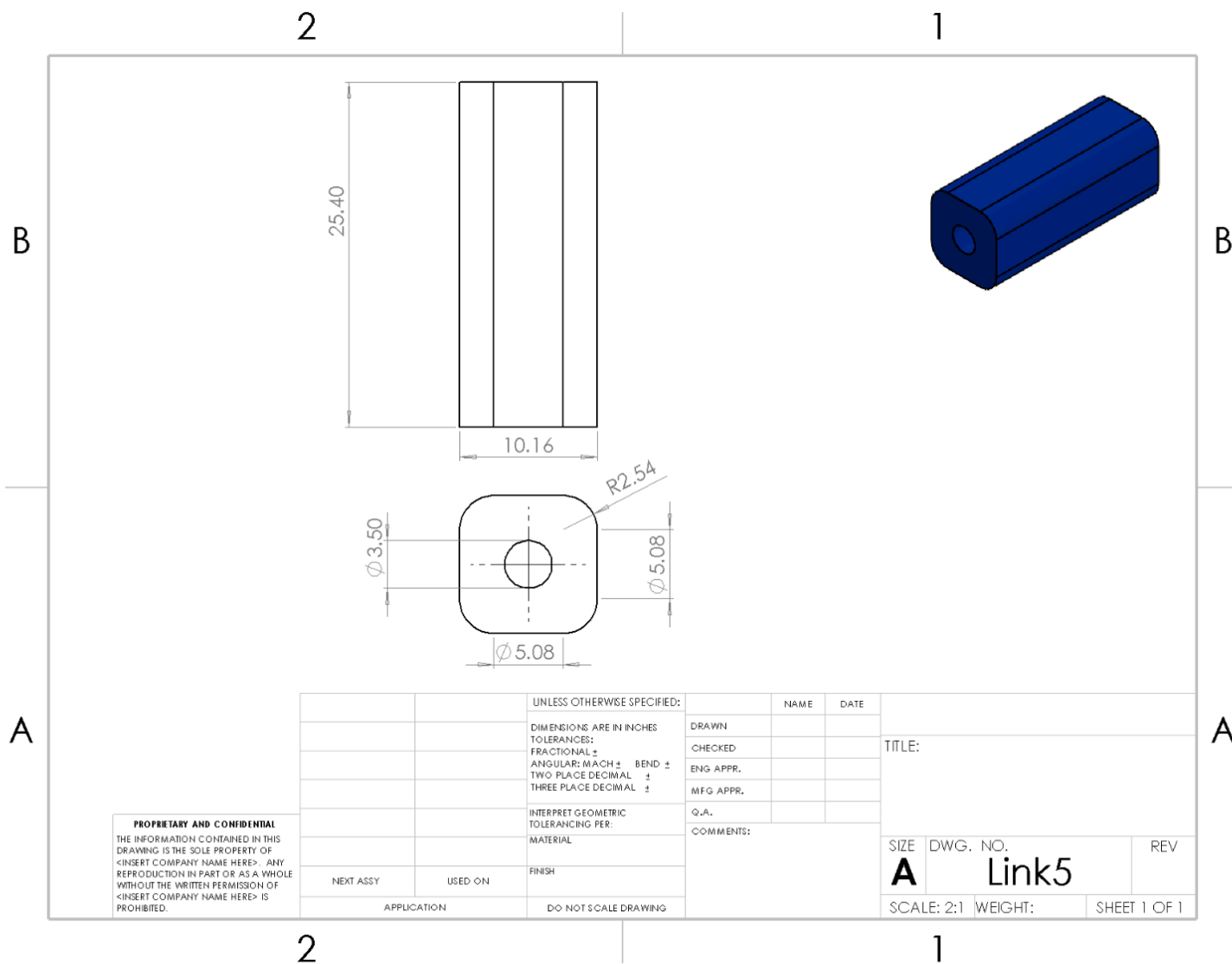
B

A

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2

1

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3.25

3.00

2.78

1.75

1.51

0.24

1.75



UNLESS OTHERWISE SPECIFIED:		NAME	DATE
DIMENSIONS ARE IN INCHES		DRAWN	
TOLERANCES:		CHECKED	
FRACTIONAL: ±		ENG APPR.	
ANGULAR: MACH ± BEND ±		MFG APPR.	
TWO PLACE DECIMAL ±		Q.A.	
THREE PLACE DECIMAL ±		COMMENTS:	
INTERPRET GEOMETRIC TOLERANCING PER:			
MATERIAL			
NEXT ASSY	USED ON	FINISH	
APPLICATION		DO NOT SCALE DRAWING	

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TITLE:

SIZE **A** DWG. NO. **Gear Guard** REV

SCALE: 1:2 WEIGHT: SHEET 1 OF 1

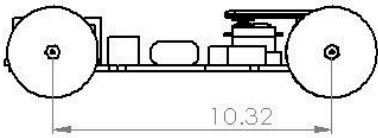
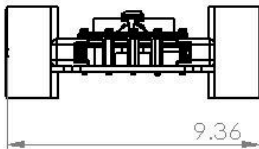
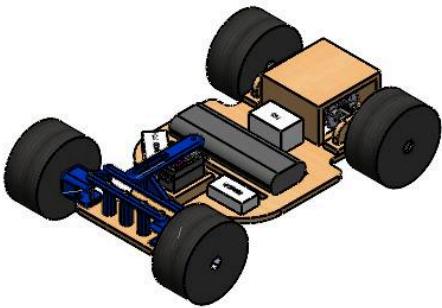
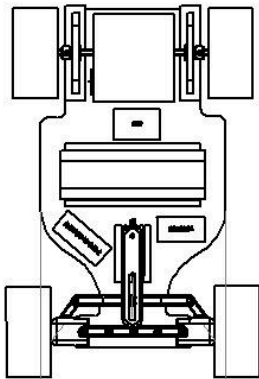
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1

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PROHIBITED.

		UNLESS OTHERWISE SPECIFIED:		NAME	DATE
		DIMENSIONS ARE IN INCHES		DRAWN	
		TOLERANCES:		CHECKED	
		FRACTIONAL $\pm$		ENG APPR.	
		ANGULAR: MACH $\pm$ BEND $\pm$		MFG APPR.	
		TWO PLACE DECIMAL $\pm$		Q.A.	
		THREE PLACE DECIMAL $\pm$		COMMENTS:	
		INTERPRET GEOMETRIC TOLERANCING PER:			
		MATERIAL			
		FINISH			
NEXT ASSY	USED ON				
APPLICATION		DO NOT SCALE DRAWING			

TITLE:

SIZE	DWG. NO.	REV
A RoverProto2		
SCALE: 1:10	WEIGHT:	SHEET 1 OF 1

Appendix B: Bill of Materials

PURCHASE ORDER FORM

The Henry Samueli School of Engineering

One PO per team. Parts will be given to you after submitting the PO. The gears and wheels will be given on a first come first served basis.

Requested By:*	Seaside Warriors	Team Captain E-Mail:*	ejchoi5@uci.edu
	Team Name		
Lab Instructor Name:*	Zarya Rajestari	Lab Section, Day, and Time:*	Wednesday, 10am
Account Name:*	ENGR 7A	Date of Request:*	10/23/24

Account*	Fund*	Sub	Project	% (If split)	Amount	Accounting Review
ENGR 7A	56123	03	7A	100%	\$225.35	
				0%	\$0.00	

Quantity*	# of Units in package*	Company*	Item Description*	Catalog/ASIN #*	Price*	Estimated Extended Price*
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example:

1	10	Amazon	Servo Horn	B07MFK266B	\$17.99	\$1.80
1	2	Amazon	Brushed ESC	B087NF55VD	\$23.79	\$11.90
1	4	Amazon	Buck Converter	B079N9BFZC	\$13.99	\$3.50
1	1	Amazon	Digital Transmitter and Receiver Radio System	B00KHJ262Y	\$75.99	\$75.99
1	2	Amazon	Battery	B07VLKP6RJ	\$31.49	\$15.75
1	2	Pololu	Motor Bracket	2676	\$7.95	\$3.98
1	2	Bilda	D shaft (50 mm length)	2101-0006-0050	\$2.49	\$4.98
1	1	Bilda	D shaft (200mm length)	2101-0006-0200	\$5.49	\$5.49
4	1	McMaster	Sleeve bearing	2938T756	\$2.02	\$8.08
8	1	McMaster	Shaft Collar	6056N14	\$2.08	\$16.64
4	2	Pololu	Wheel Adapter	2686	\$4.95	\$9.90
1	1	Servocity	Pinion Gear	2303-0004-0015	\$9.99	\$9.99
1	1	Servocity	Gear Hub	1309-0016-0006	\$7.99	\$7.99
1	1	Servocity	Hub-Mount Gear (52T)	2312-0414-0052	\$3.39	\$3.39
4	4	Amazon	85 mm Wheels	B07BS9QJ8L	\$16.99	\$16.99
2.5	1	N/A	3D Printing	N/A	\$4.00	\$10.00
1	1	N/A	6mm italian poplar plywood 12"x12"	N/A	\$2.78	\$2.78
					Subtotal	\$209.14
					*Tax rate: 7.75%	\$16.21
					TOTAL ORDER PRICE	\$225.35

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<https://www.americanspecialops.com/vehicles/sov/>
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